

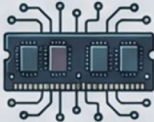
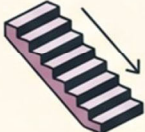
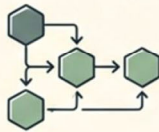


MCH R User Group – Intro to R

Jason Geslois, MPH, MAS, EMBA
Senior Epidemiologist

MCH R User Group – Intro to R

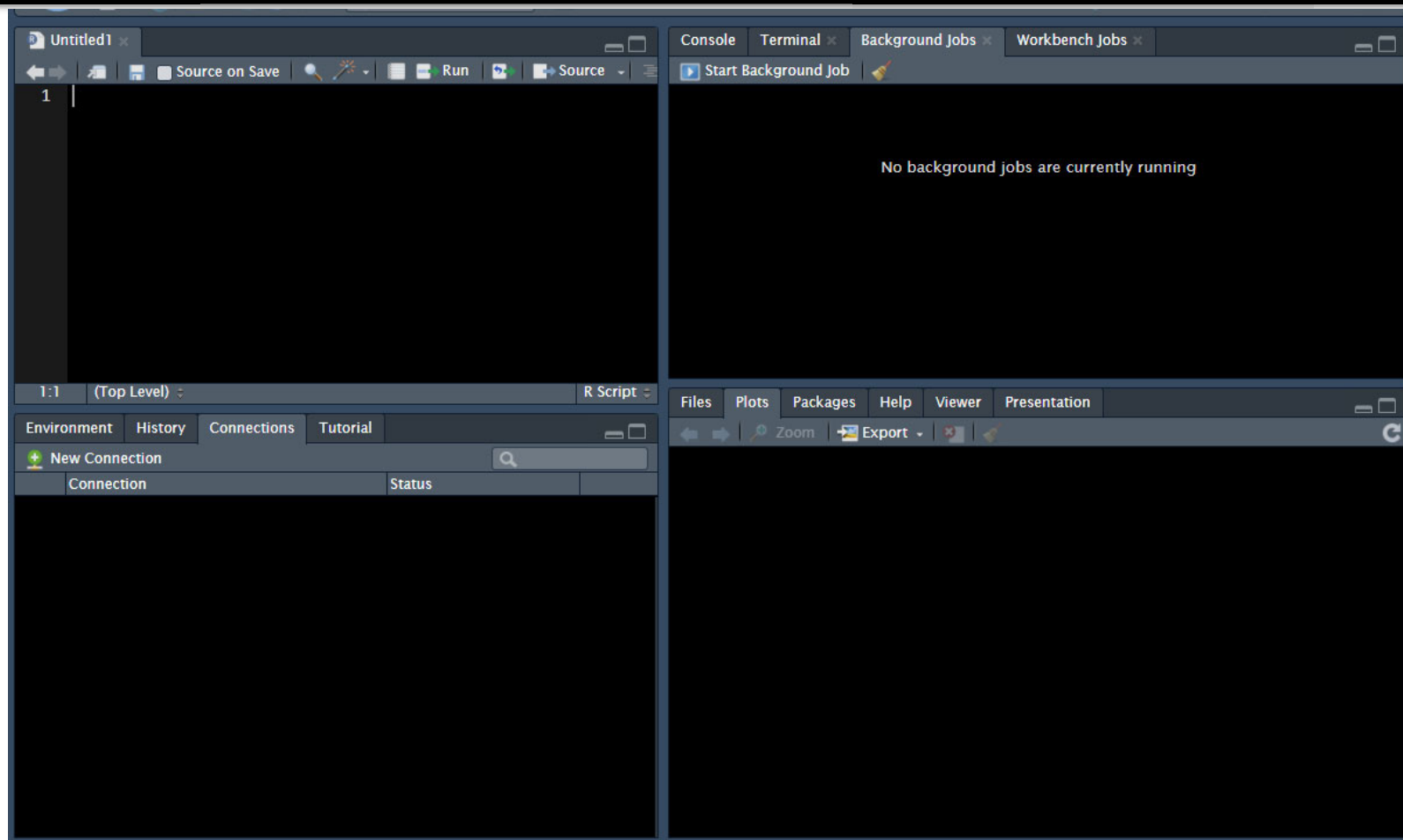
Translating the Philosophy: SAS vs. R

Dimension	Legacy (SAS)	Modern (R)
Data Storage	 Disk-based tables. Persistent storage.	 In-memory Data Frames. Active RAM.
Execution	 Sequential PROCs & DATA steps.	 Interactive, object-oriented functions.
Ecosystem	 Proprietary, built-in modules.	 Open-source, community Packages.
Workspace	 Libraries and Libnames.	 Self-contained RStudio Projects.

NotebookLM

To protect and improve the health and environment of all Kansans

MCH R User Group – Intro to R



To protect and improve the health and environment of all Kansans

MCH R User Group – Intro to R

The Command Center

 →  Translation Key

Source / Files

The Drafting Board.

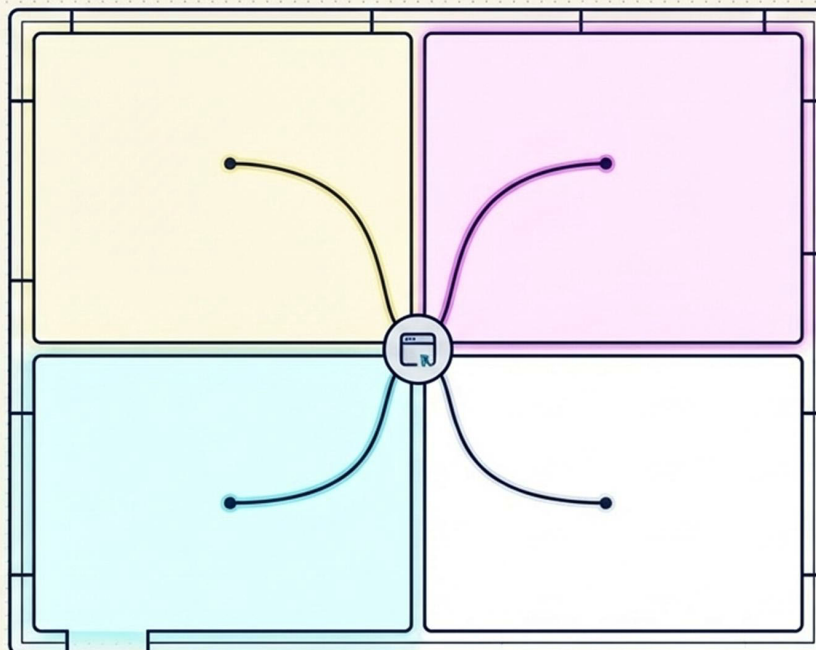
Where scripts, markdown, and quarto files are written.

Console

The Engine. Where code executes instantly.



Unlike SAS, you don't need a Run command for every line; the console is live.



Environment

Active Memory.

Displays loaded data and saved objects.

Output

The Display. Houses the Plot pane, Help pane, and local file directory.

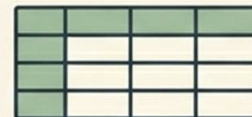
MCH R User Group – Intro to R

The Drafting Table (Source)



Where you write reproducible code.

The Workbench (Environment)



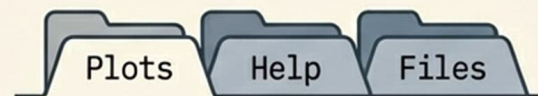
View active objects, loaded Data Frames, and manage memory.

The Engine Room (Console)



For immediate execution and testing code on the fly.

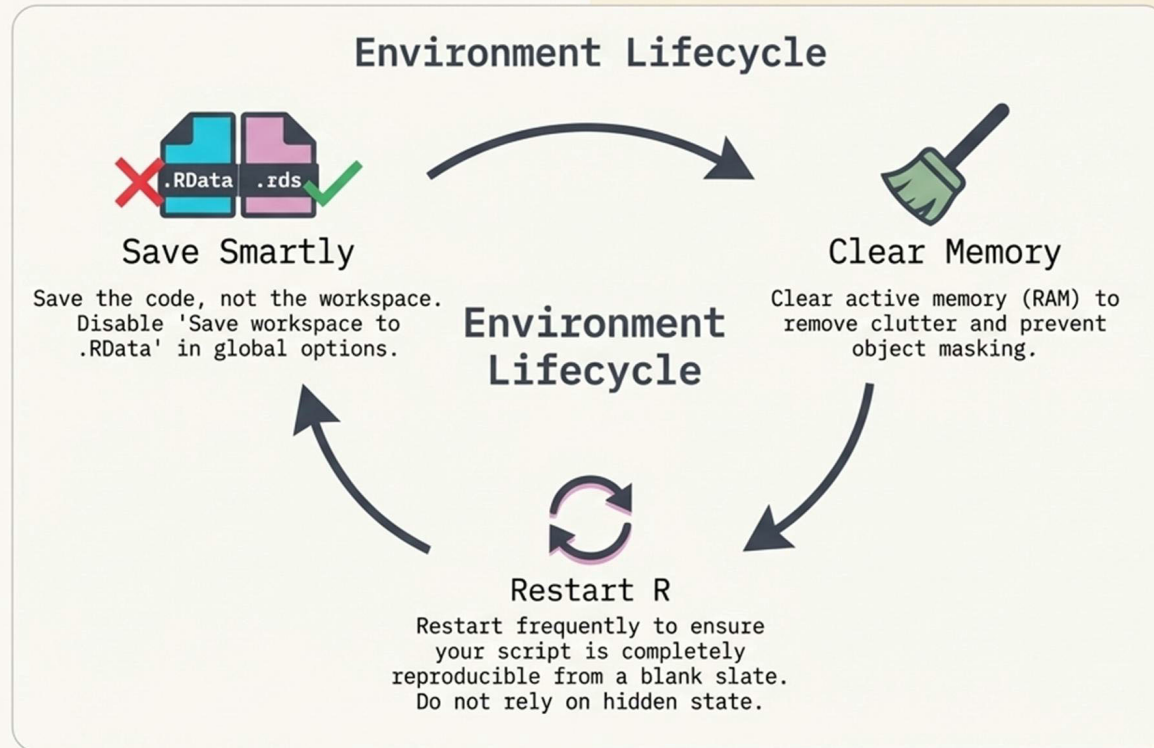
The Display (Output)



Review visualizations, read package documentation, and browse your directory.

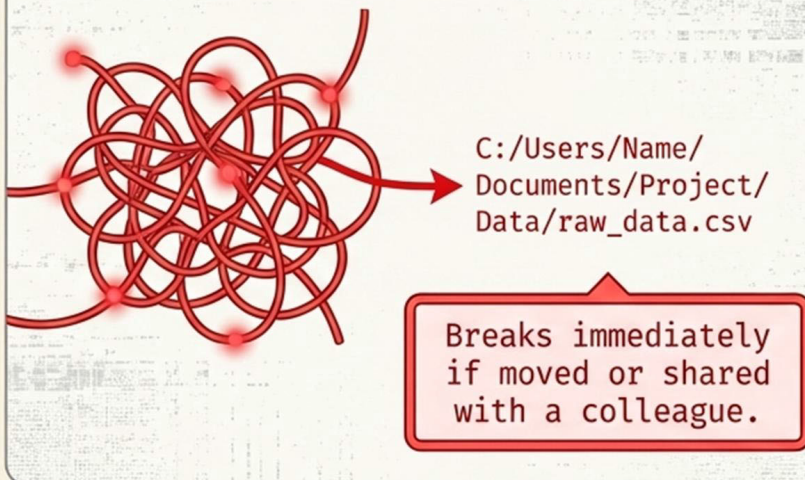
MCH R User Group – Intro to R

Session Management and Active Memory

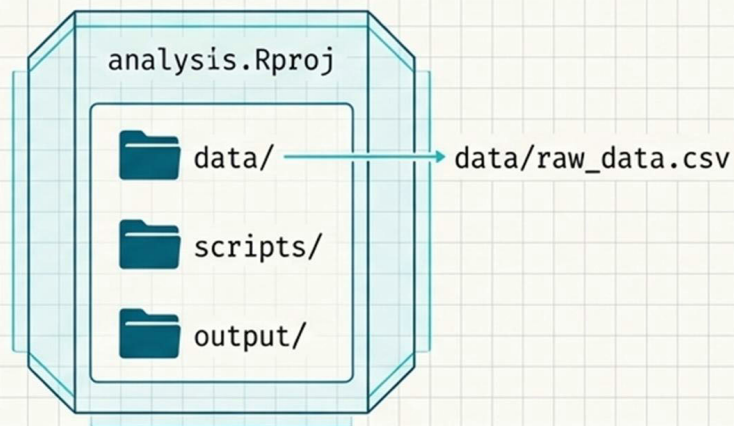


MCH R User Group – Intro to R

The Old Way



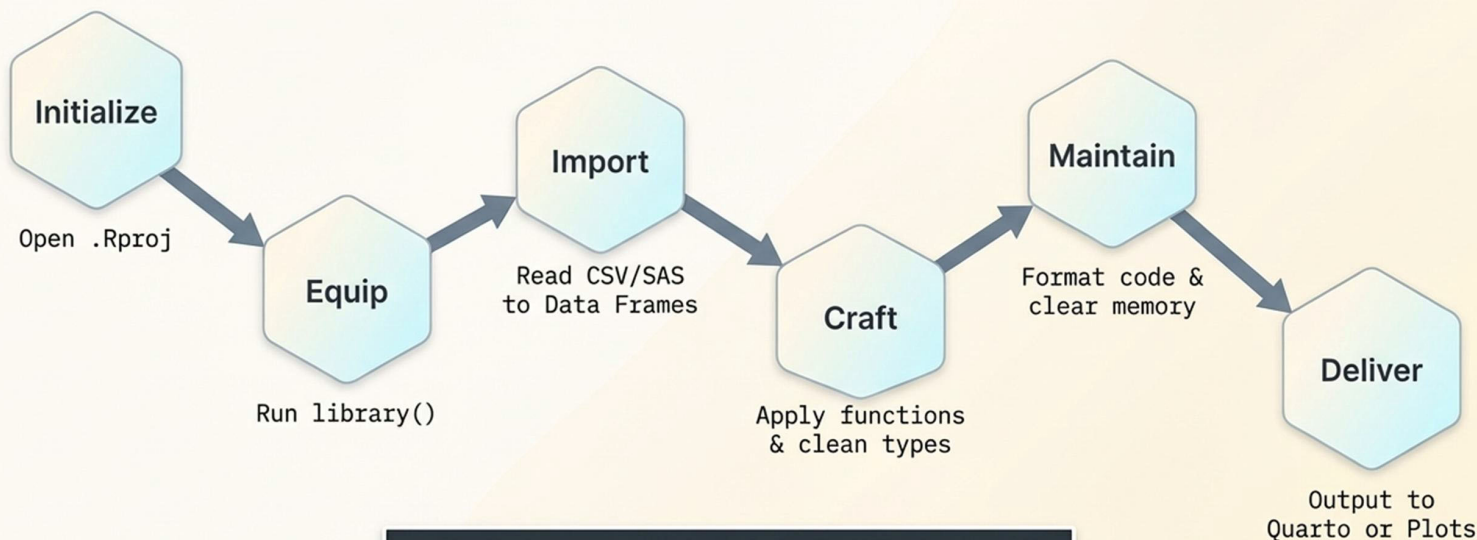
The R Project Way



A Project is a self-contained universe. By using relative paths, the code just works—no matter whose computer opens it.

MCH R User Group – Intro to R

Synthesis: The R Workflow Blueprint



You now possess the blueprint. The transition from legacy scripts to reproducible environments begins here.

MCH R User Group – Intro to R

Document Decision Matrix

Scripts (.R)

Use Case: Backend data processing, simple functions.

Analogy: The rough draft / engine room.



Markdown (.Rmd)

Use Case: Interleaving text, code, and plot panes.

Analogy: The standard analyst notebook.



Quarto (.qmd)

Use Case: Modern, multi-format publishing (HTML, PDF, slides).

Analogy: The final polished report.



The Modern Standard

MCH R User Group – Intro to R

```
{r setup, include=FALSE}  
knitr::opts_chunk$set(echo = TRUE)
```



Loading Libraries & Data Directories

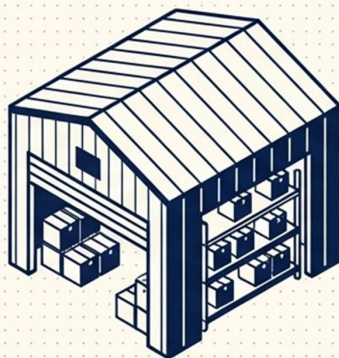


Here is where you first load the packages.

The Package Supply Chain

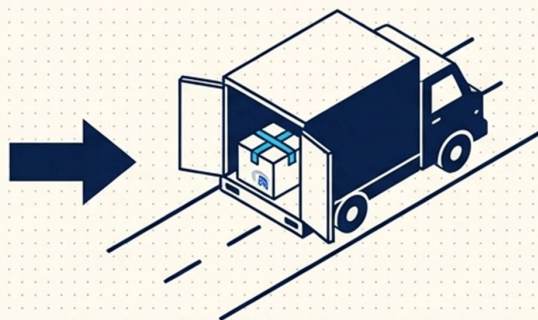
 SAS to R Translation Key 

CRAN (The Warehouse)



The global repository of open-source packages.

`install.packages()`



 **The Delivery:** Do this only once per machine to bring the tool to your hard drive.

`library()`



 **The Workbench:** Take the tool out of your local library and put it on your active desk. Do this every time you start a new session.

 NotebookLM

MCH R User Group – Intro to R

Rainbow Code

```
(function(data){
  return data.filter(d =>
    (d.value > 10 &&
      (d.type === 'A' ||
        (d.type === 'B' && d.status === 'active'))));
})
```

Rainbow brackets prevent missing parenthesis errors and visually untangle complex nesting.

Find / Replace All

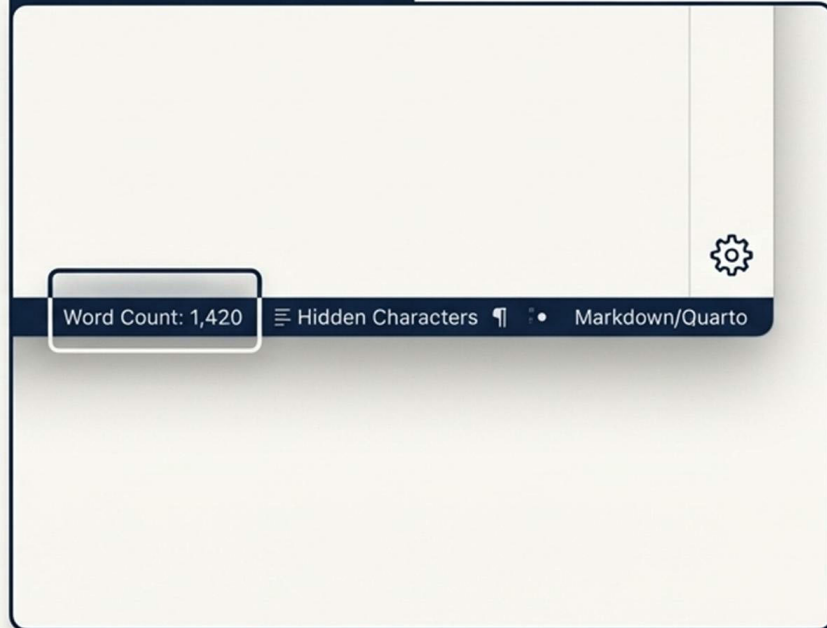
```
// unuse script sonable
for (variable, source's old_variable) = new_clean_var {
  return return, new_clean_var => old_variable;
}

//
if (new_variable = as.character) {
  resolve(new_clean_var variable, new_clean_var => new_clean_var);
}
```

Find: old_variable → Replace: new_clean_var

Instantly transform variable names across a whole script without hunting.

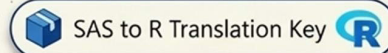
Status Bar & Global Options



Word Count: 1,420 Line: 87, Col: 15

Enable word counts and hidden character views to optimize writing in Markdown and Quarto.

Code Craftsmanship



The Messy Block

```
data mess; set raw;
  if x=1 then do;
    y=x*2;
    z=y+3;
    if z>10 then w=z-2; else w=z+2;
  end;
  else do;
    y=x*3;
    z=y+2; if z<5 then w=z/2;
  else w=z*2; end; run;
proc print data=mess; run;
```

The Clean Block

```
# Data Transformation for Final Output
messy_data <-
  raw_data |>
  mutate(
    y = case_when(
      x == 1 ~ x * 2,
      TRUE ~ x * 3
    ),
    z = case_when(
      x == 1 ~ y + 3,
      TRUE ~ y + 2
    ),
    w = case_when(
      z > 10 ~ z - 2,
      z < 5 ~ z / 2,
      TRUE ~ z + 2
    )
  )
```

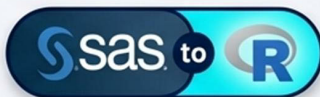
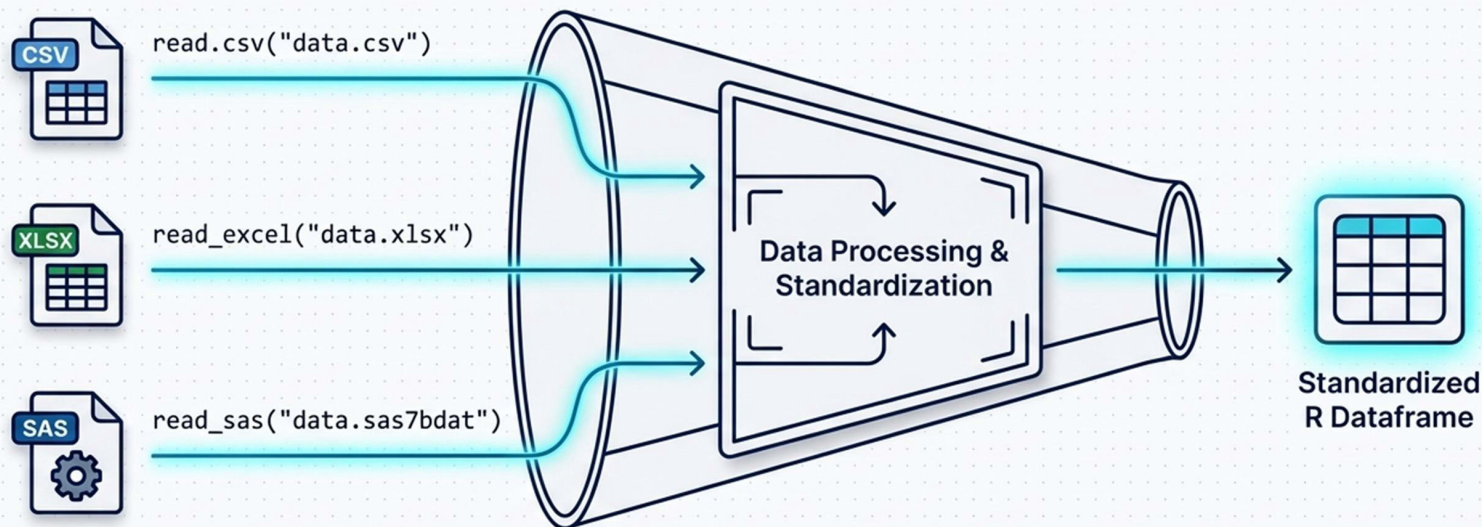
Commenting: Use # for inline context and to divide code into functional sections.

Reflow: Wrapping long text lines for readability.

Re-indent: Automatically aligning nested logic.

MCH R User Group – Intro to R

Data Ingestion Pipeline



SAS to R
Translation Key

In SAS, you write LIBNAME statements.
In R, you assign reading functions directly to an object.

Anatomy of R Syntax



SAS

```
PROC MEANS DATA=mydata; VAR myvar; RUN;
```

R

```
summary_data <- calculate_mean(data, na.rm = TRUE)
```

The Environment Object

This pushes
the result into
memory.

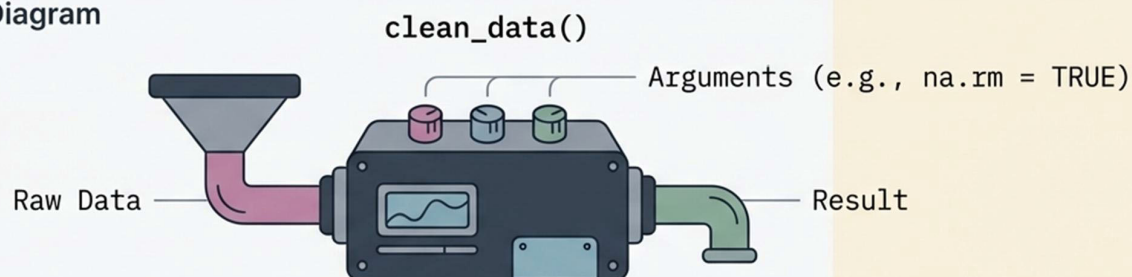
The Assignment
Operator

The Simple Function

The Arguments
(controlling function behavior)

Crafting with Functions

The Blueprint Diagram



Code Comparison

Run 1

```
mean(sales_data, na.rm = FALSE)
```

NA

Run 2

```
mean(sales_data, na.rm = TRUE)
```

42.5

Highlight and run different lines of code sequentially in the Console to instantly compare outputs and tweak function arguments.

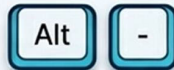
MCH R User Group – Intro to R

Quick Reference / Cheat Sheet

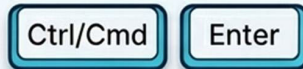


Top Keyboard Shortcuts

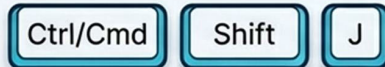
Assignment (<-)



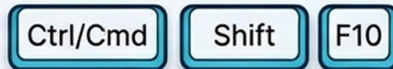
Run Code



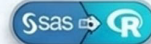
Find/Replace All



Restart Session



SAS to R Rosetta Stone Summary



Workspace



R Project

LIBNAME



read_*() functions

PROC



Modular Functions

.sas



.R, .Rmd, .qmd

MCH R User Group – Intro to R

Keyboard Shortcut Quick Reference

Source on Save

Run

Source

Tabs

Ctrl+Shift+.

Switch to Tab...

Ctrl+F12

Next Tab

Ctrl+F11

Previous Tab

Ctrl+Shift+F11

First Tab

Ctrl+Shift+F12

Last Tab

Panes

Ctrl+1

Move Focus to Source

Ctrl+2

Move Focus to Console

Ctrl+3

Move Focus to Help

Ctrl+4

Show History

Ctrl+5

Show Files

Ctrl+6

Show Plots

Ctrl+7

Show Packages

Ctrl+8

Show Environment

Ctrl+9

Show Viewer

Ctrl+F1

Show Vcs

Ctrl+F2

Show Build

Ctrl+F5

Show Connections

Ctrl+F6

Show Presentation

Ctrl+F7

Add Source Column

Ctrl+Shift+0

Show All Panes

Add Shift to zoom (maximize) pane.

Files

Ctrl+S

Save

Ctrl+Alt+S

Save All

Ctrl+Alt+Shift+N

New R Script

Ctrl+O

Open File...

Ctrl+W

Close

Ctrl+Alt+Shift+W

Close All Except Current

Ctrl+Shift+W

Close All

Ctrl+Shift+F

Find in Files...

Source Navigation

Ctrl+F9

Back

Ctrl+F10

Forward

Ctrl+Alt+U

Find Usages

Ctrl+F3

Use Selection for Find

Ctrl+F

Find...

F3

Find Next

Shift+F3

Find Previous

Ctrl+Shift+J

Replace and Find

Ctrl+.

Go To File/Function...

Alt+Shift+G

Go to Line...

Alt+Shift+J

Jump To...

Ctrl+P

Jump To Matching

Ctrl+Alt+Shift+E

Expand To Matching

Ctrl+Shift+O

Show Document Outline

Ctrl+Alt+Up

Add Cursor Above Current Cursor

Ctrl+Alt+Down

Add Cursor Below Current Cursor

Ctrl+Shift+Up

Expand Selection

Ctrl+Shift+Down

Shrink Selection

Ctrl+PgDn

Go to Next Section

Ctrl+PgUp

Go to Previous Section

Ctrl+Alt+A

Split Into Lines

Ctrl+Alt+Shift+A

Edit Lines from Start

Ctrl+Shift+F4

Use Visual Editor

Ctrl+Alt+Shift+Up

Move active cursor up

Ctrl+Alt+Shift+Down

Move active cursor down

Execute

Ctrl+Shift+S

Source Active File

Ctrl+Shift+Enter

Source with Echo

Ctrl+Shift+Enter

Preview JS

Ctrl+Shift+Enter

Preview SQL

Ctrl+Alt+G

Source a File...

Ctrl+Alt+P

Re-Run Previous

Ctrl+Enter

Run Selected Line(s)

Alt+Enter

Run Line(s) without moving cursor

Ctrl+Alt+R

Run All

Source Editor

Ctrl+Alt+I

Insert Chunk

Ctrl+Shift+R

Insert Section...

Ctrl+Alt+X

Extract Function

Ctrl+Alt+V

Extract Variable

Ctrl+Shift+C

Comment/Uncomment Lines

Ctrl+I

Reindent Lines

Ctrl+Shift+/

Reflow Comment

Ctrl+Shift+A

Reformat Selection

Ctrl+Alt+Shift+B

Show Diagnostics (Project)

Alt+L

Collapse Fold

Alt+Shift+L

Expand Fold

Alt+O

Collapse All Folds

Alt+Shift+O

Expand All Folds

Alt+Up

Move Lines Up

Alt+Down

Move Lines Down

Ctrl+D

Delete Line

Ctrl+U

Yank Line Up to Cursor

Ctrl+K

Yank Line After Cursor

Ctrl+Y

Insert Yanked Text

Alt+-

Insert Assignment Operator

Ctrl+Shift+M

Insert Pipe Operator

Ctrl+Alt+Shift+M

Rename in Scope

Ctrl+Alt+Shift+R

Insert Roxygen Skeleton

Ctrl+Backslash

Request Copilot Code Suggestions

Debug

Shift+F9

Toggle Breakpoint

F10

Execute Next Line

Shift+F4

Step Into Function

Shift+F7

Finish Function/Loop

Shift+F5

Continue

Shift+F8

Stop Debugging

Accessibility

Alt+Shift+V

Screen Reader Support

Source Control

Ctrl+Alt+D

Diff Files

Ctrl+Alt+M

Commit...

Build

Ctrl+Shift+K

Compile PDF

Ctrl+Shift+K

Preview

Ctrl+Shift+K

Knit Document

Ctrl+Shift+K

Render Document

Ctrl+Shift+K

Run Document

Ctrl+Shift+B

Install Package

Ctrl+Shift+L

Load All

Ctrl+Shift+E

Check Package

Ctrl+Alt+F7

Test Package

Ctrl+Shift+D

Document

Console

Ctrl+L

Clear Console

Ctrl+Up

Popup Command History

Terminal

Alt+Shift+R

New Terminal

Alt+Shift+M

Move Focus to Terminal

Alt+Shift+F11

Previous Terminal

Alt+Shift+F12

Next Terminal

Other

F1

Show Function Help

F2

Go To Function / File

Ctrl+Space

Complete Code

Ctrl+Shift+F10

Restart R

Ctrl+Alt+F11

Previous Plot

Ctrl+Alt+F12

Next Plot

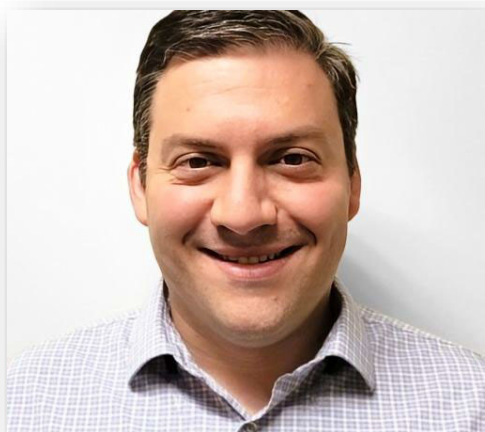
Ctrl+Shift+*

Log focused element

RStudio: Posit Software PBC

To protect and improve the health and environment of all Kansans

Now for a Live Demo



Jason Geslois

Senior Epidemiologist

Kansas Department of Health and Environment

Jason.geslois@ks.gov



kdhe.ks.gov



@KDHEnews



@kdheks



@KDHE



@Kansas Department of Health and Environment

Thank You/Questions

